Silent Acceptance: A Falsification-Side Benchmark of Form-Specification Gates for Physical Model Discovery

Complete draft v3 — working manuscript, not submitted. Supersedes `draft_v2_complete.md`. Changes from v2: two suites are added — **S5, the gate-comparison suite** (the benchmark's most obvious attack, "why not just compare against a nested rival?", answered with data; `runs/gate_cmp`; 20,400 datasets) and **S6, a real-data suite** on World Bank open macro data (`runs/wb_real`; 165 datasets, external validity + a live dilution signature) — and the paper is **repositioned** accordingly: no detector is claimed as the gate; the deliverable is a measured failure-mode map plus a deployment ordering. All v2 numbers are retained as audited; the change notes, including both revision rounds, are Appendix G.

Abstract

Symbolic regression and physical model discovery pipelines certify candidate models with goodness-of-fit metrics — R^2, cross-validation error, information criteria, coefficient standard errors — that measure how well a model fits, never whether an alternative model family would fit strictly better. A model can score near-perfectly on all of these metrics while having the wrong functional form, and fail catastrophically the moment it extrapolates. Classical form-specification tests exist — Ramsey's RESET and its variants, with heteroskedasticity-robust inference — and the still-simpler workflow alternative exists too: fit a nested rival and compare. To our knowledge, no benchmark measures whether any of these gates work on the data distributions such pipelines actually consume: physical laws sampled over physically realistic ranges with multiplicative noise, evaluated in log-transformed space — and none measures what happens on real observational data.

We introduce a falsification-side benchmark filling this gap. Six suites total 1,480 grid cells and 508,825 fitted datasets, with 472,500 further held-out evaluations: (S1) 19 controlled synthetic scenarios; (S2) 32 tasks from the SRSD-Feynman registry under official sampling configurations; (S3) 30 Feynman equations stratified by structural family; (S4) a 188-cell family-viability battery extending the same machinery from rejection to routing; (S5) a gate comparison placing the benchmark's best test against the "obvious alternative" it had never faced — a nested quadratic rival adjudicated by 5-fold cross-validation or AICc; and (S6) six macroeconomic relations from World Bank open data across four years and ~190 economies per cell. All methods run under one protocol: a common significance level; power compared only at matched empirical false-positive rate; preregistered labels; explicit registration of every degenerate or rank-deficient evaluation.

Five findings stand out. **First**, the phenomenon is real and silent: a power law fitted to Prandtl–Karman friction data certifies itself at $R^2 = 0.99$ through every metric a pipeline computes, while extrapolating at −20% one decade out. **Second**, the benchmark caught — and its revision retracted — its own most striking claim: our harness's HC3-Wald path rejects at rate 1.00 on 5 of 13 physics-derived tasks whose built designs are near-collinear (pooled 42–56% false alarms, flat in n), but an independent SVD-based implementation (statsmodels) is nominal-compatible (.067, Wilson [.034,.126]) on the same data and agrees with ours to three digits elsewhere: the failure is a silent numerical pathology of pinv-squared-conditioning inference paths, not of the robustness correction — a deployment hazard discovery code should test for. **Third**, no test is uniformly best; rankings invert under matched false-positive rates and again across task families, and each method's strength sits on another's blind spot. **Fourth**, the battery extends rejection to routing after its null is rebuilt per family — null behavior is coordinate-dependent (the residual permutation is empirically calibrated in log-log coordinates on every pool we measured; 30–34% false-escalation in rational coordinates until a wild bootstrap repairs it). **Fifth** — the benchmark's most uncomfortable results — the nested quadratic rival is the better default detector wherever a reference pool exists, and the naive OR union of the permutation gate with the CV rival, previously the best-recall cell, owes that advantage to a larger joint false-positive budget: recalibrated to a 5% joint budget, the union matches but does not beat AICc-quad. On real macroeconomic data the gates agree where signal is clear, diverge where the map predicts weakness, and show an observational pattern consistent with the synthetic dilution signature. The deployment ordering: default AICc-quad when a reference pool is available; the permutation gate when a self-calibrating p-value is required; recall-critical deployments should budget for AICc-quad, not for an uncalibrated union. We release all generators, per-cell results, failure registrations, and content-hashed configurations under fixed seed rules.

1 Introduction

A regression model can be simultaneously excellent on every certification metric a discovery pipeline computes and wrong in its functional form. A power law fitted to Prandtl–Karman friction data reaches $R^2 = 0.99$, passes k-fold cross-validation, wins its information-criterion comparison, and reports small standard errors — while extrapolating with −20% error one decade outside the training domain. This is not hypothetical: the failure is reproduced as a controlled scenario in our laboratory notebook and appears in this benchmark as scenario S1 (Section 3.3), where all four specification tests detect it at power ≥ 0.99. Fit quality and form correctness are different axes, and current discovery practice measures only the first.

The statistics literature has owned the second axis for over fifty years. Ramsey's RESET [1] tests whether powers of the fitted values carry additional signal; built-regressor variants use powers of the regressors; White [2] and MacKinnon–White [3] provide heteroskedasticity-robust inference. And the workflow alternative that requires no new statistics at all — fit a nested rival family and let cross-validation or an information criterion arbitrate — has been standard practice in applied regression for as long. Yet none of this is measured by symbolic regression benchmarking. SRBench [5] measures whether discovery methods find the right expression across 252 problems; SRSD [6] redesigns Feynman-derived tasks with realistic sampling ranges. In neither framework does anything measure whether an attached falsification layer would reject a wrong form. The asymmetry has practical consequences: a discovery pipeline with no working falsification layer silently accepts misfit forms, and its downstream use — extrapolation, engineering design, scientific claim-making — inherits the error silently.

Measuring the falsification side faithfully requires resolving a protocol problem that, to our knowledge, no prior work addresses for this test family: the candidate gates reject on incommensurable bases. A bootstrap-t threshold ("alarm when $|t| > 3$"), an asymptotic F p-value ("reject when $p < 0.05$"), and a cross-validated RMSE difference ("escalate when the rival wins") measure different things; comparing them directly measures threshold placement, not detection. Our first pilot compared a native $|t| > 3$ alarm against $p < 0.05$ and produced inverted rankings that vanished under a common basis. The protocol below (Section 3.1) rests on three commitments: one decision basis per comparison; power compared only at a matched empirical false-positive rate estimated on true-monomial tasks; and explicit registration of every failure.

This paper asks its benchmark the three questions a reviewer would ask, in the order they would ask them. (i) *Do the classical tests work at all on physical data distributions?* — Sections 4.1–4.5: calibration, power, blind zones, extrapolation semantics. (ii) *A rejection terminates inquiry — then what?* — Section 4.6: a family-viability battery that routes instead of refusing, once its null is rebuilt per coordinates. (iii) *Why not just fit a nested rival and compare?* — Section 4.7: because until it is measured, that question has no answer; measured, it wins, and the benchmark says so. A real-data suite (Section 4.8) then asks the external-validity question: does the synthetic failure-mode map predict anything about live observational data? It does, including one live reproduction of a synthetic blind spot.

The question this benchmark answers is narrow and falsifiable: **when the ground truth is known and the candidate form is wrong, which gates detect the misspecification — at what false-alarm cost, on what reference budget, and do the synthetic failure modes transfer to real data?** We contribute: (i) the benchmark — six suites, 508,825 fitted datasets plus 472,500 held-out evaluations, released generators with seed rules making any cell independently reproducible; (ii) the protocol — matched-basis comparison with empirical-FPR calibration and failure registration; (iii) the failure-mode map — including a silent numerical failure of pinv-based HC3-Wald inference on near-collinear physics-derived designs (independently verified; the correction itself is not implicated), the near-blindness of fitted-F RESET to rational laws, and the measured dominance and calibration cost of the nested-rival gates; (iv) the family-viability battery — 94,000 datasets showing rejection extends to routing once — and only once — the null is rebuilt per coordinates; and (v) a real-data suite showing the map's mechanisms operating outside the laboratory. What we deliberately do not contribute is a winning detector; the benchmark's own results (Section 4.7) demote its best test from default gate to portfolio member, and the paper is organized around that honesty.

2 Related Work

Specification testing. RESET [1] augments the regression with powers of the fitted values and tests the joint significance of the added coefficients; the exog variant uses powers of the regressors instead. Heteroskedasticity-consistent covariance estimation [2] and its HC3 leverage-weighted refinement [3, 4] are the standard small-sample corrections and are the default in several mainstream econometrics and statistics packages.

These results are derived for linear models under fixed or quasi-fixed designs; none of the classical treatments contemplates log-transformed physical laws whose regressors span only a few orders of magnitude — precisely the regime our benchmark shows breaks HC3's small-sample behavior (Section 4.1).

Model selection as a specification gate. Comparing a null model against a nested rival by information criterion [12] or cross-validation [13] is the oldest workflow-level misspecification check in applied practice. It is so obvious that a reviewer of any new test will ask for it; it is so workflow-dependent (which rival? which criterion? which threshold?) that benchmarking it requires exactly the protocol this paper introduces — a frozen rival family, frozen decision statistics, and thresholds calibrated on a reference pool. Suite S5 is, to our knowledge, the first head-to-head we know of between a resampling-null specification test and nested-rival selection gates under matched false-positive rates.

Resampling nulls under non-i.i.d. errors. The wild bootstrap [10, 11] was developed for inference under heteroskedasticity, and its permutation cousin requires exchangeability. The battery's mechanism finding (Section 4.6) is a compact new instance of this classical dichotomy: a log transform homoskedasticizes multiplicative noise, so residual permutation is valid in coordinates ending in log y and invalid in coordinates ending in y or 1/y — the null's validity inherits the noise geometry of the coordinates the test runs in.

Symbolic regression benchmarks. SRBench [5] established the discovery-side evaluation: 14 SR methods and 7 ML baselines on 252 problems, scored by exact recovery and held-out accuracy. SRSD [6] reconceived the task distribution itself — realistic per-variable ranges reviewed against the physics — and is the source of our task configurations. Both measure the discovery side; a pipeline that returns a wrong expression and one that flags it score identically. Our generators deliberately reuse SRSD's official sampling objects so that the falsification-side pools are the same distributions the discovery side already standardizes on.

Physics-informed and dimensionally-aware discovery. AI Feynman [7] uses physical units to constrain the hypothesis space; measurement-driven discovery distills governing equations and structure directly from data [8]; PySR [9] represents the modern search side. Dimensional correctness is necessary but not sufficient for form correctness: every ground-truth law in our suites is dimensionally self-consistent, yet each non-monomial law is a form misspecification for a power-law model. The falsification side complements these constraints — units narrow what to search; specification gates check what was found.

Honest-failure evaluation. Existing SR benchmarks report no analogue of a false-positive rate because they contain no explicit rejection mechanism to calibrate. We are not aware of a symbolic regression benchmark that measures attached falsification behavior, per family, with preregistered labels and failure registration — or one that benchmarks its own proposed test against the nested-rival alternative its reviewers would name. This is the gap the present benchmark occupies.

The wider specification-testing literature. Econometrics and statistics hold many more form-specification tools than the gates tested here: smooth residual tests, generalized likelihood-ratio and conditional moment tests, rainbow and related one-degree-of-fitting checks, spline/GAM residual comparison, kernel-based lack-of-fit statistics, cross-fitted and sample-splitting variants, and conformal-style selective-risk diagnostics. S5's nested-quadratic rival is the discovery community's canonical objection, not a survey of this literature; LOOCV, BIC, held-out likelihood, or non-nested rivals may rank differently. The gates were chosen because they are what discovery pipelines actually attach today (RESET variants, robust inference) plus the workflow alternative a reviewer would name (nested-rival selection); the harness is gate-agnostic — a new gate is one statistic-plus-decision function over the same fitted cells — and extending the benchmark to these tests is future work, not a closed set.

3 Benchmark Design

3.1 Design principles

P1 — Common rejection basis. Within any comparison, all gates decide on the same basis at the same significance level $\alpha = 0.05$. The permutation curvature test's empirical p-value is computed from $B = 999$ permutations of the residual-permutation null; RESET variants use asymptotic F reference distributions; S5's selection gates use thresholded statistics, with thresholds at naive zero and at an H0-pool-calibrated value — and naive versus calibrated results are reported side by side, because raw tables are what current practice would see.

P2 — Power only at matched empirical FPR. Headline power comparisons calibrate each method's decision threshold to a 5% empirical false-positive rate, estimated on pooled true-monomial tasks within the same suite,

sample-size stratum, and noise regime.

P3 — Preregistered labels. H0 (strict monomial: the log-log mean function is exactly linear) versus H1 (any non-monomial structure) is fixed from symbolic structure of the ground truth before any test runs. The family stratification of S3, the battery's truth registry, the S5 gate definitions, and the S6 relation list are likewise frozen in code before their runs; no label or configuration is revised after seeing results.

P4 — Explicit failure registration. Every evaluation that cannot produce a valid decision is registered with a typed reason — zero-residual degeneracy, design rank deficiency, insufficient residual degrees of freedom, leverage degeneracy, singular covariance, statistic mismatch, generation failure (S5 SRSD side), indicator archival (S6) — and excluded from rate numerators while remaining in the denominators' audit trail. The zero-noise grid cells exist to exercise this path.

P5 — Official data configurations. S2/S3 tasks reuse the official per-variable sampling objects of the source registry; tasks whose inputs or outputs are not strictly positive under the official configuration are marked not-applicable and excluded — we do not shift or rescale data to make them log-fittable, since that changes the hypothesis under test. S6 uses a public observational source (World Bank Open Data API [14]) with the relation list frozen before any gate ran; the United Nations Foundation itself publishes no dataset API, and the World Bank is the UN-system institution with the open macro data of record.

3.2 Gates under test

All gates operate in log-log space on positive data: let $ly = \log y$ and $X = [1, lx_1, \ldots, lx_k]$ be the design of the candidate power-law model.

A. Permutation curvature (form_check). Adds a single mean-centered quadratic column $q = \text{mean}_j((lx_j - \overline{lx_j})^2)$ and takes the added coefficient c_2 as statistic; size evidence comes from B = 999 residual permutations (vectorized through the pseudo-inverse row), with a statistic-mismatch sentinel comparing wrapper and native values on every call. Validity note: permuting *estimated* residuals assumes they are exchangeable, which OLS residuals are not exactly — the permutation null here is an approximation whose quality is an empirical question, not a finite-sample exactness theorem. What the benchmark establishes is calibration of this construction on our H0 pools (Tables 1–2); log-transformed multiplicative noise and favorable designs plausibly help, and §6 lists the stress tests (leverage-adjusted residuals, wild and block nulls, cluster dependence, fixed-design variants) that would bound where the approximation breaks.

B. RESET fitted, classical F. Adds $\hat{y}^2$ and $\hat{y}^3$; classical F test of joint significance.

C. RESET fitted, HC3-Wald. Same augmented design; HC3 sandwich Wald statistic on the added coefficients, referred to $F(q, n - k)$.

D. RESET exog, HC3-Wald. Built regressors are powers of each log-regressor; HC3-Wald as in C.

E. CV-quad (S5). The "obvious alternative": rival = full quadratic expansion of the log features (linear terms, per-feature squares, pairwise cross-terms; minimum-norm least squares absorbs rank deficiency). Statistic D = RMSE_cv(power) − RMSE_cv(quad) over 5 deterministic folds; escalate if $D > \delta$. Naive $\delta = 0$; calibrated δ^* chosen on the H0 pool to FPR = 5%.

F. AICc-quad (S5). Same rival; escalate if AICc(power) − AICc(quad) > m, k = column rank. Naive m = 0; calibrated m^* as in E.

Configurations were frozen before all main runs — powers (2, 3); B = 999; $\alpha = 0.05$; the rival family and criteria of E/F; the grids below — and are identical across suites. The same harness invokes all gates on every dataset, so comparisons share data, seeds, and failure handling exactly.

3.3 Task suites S1–S3

S1 — Synthetic control (19 scenarios × 500 seeds). Three monomial H0 scenarios (univariate; two-variable; two-variable with $\rho \approx 0.9$ correlated design). Eleven structural H1 scenarios: quadratic curvature ($c = 0.10$); cubic (standardized odd term); multiplicative interaction (log-space cross term); threshold (exponent 0.4 → 1.0 at the domain midpoint); exponential law; logarithmic law; rational (saturating) law; omitted variable; dilution (one of two variables curved); opposing curvature (+0.12, −0.12 on an exchangeable design, so the mean-square column's population coefficient is exactly zero); collinear-quadratic ($\rho \approx 0.95$ design with curvature). Five noise-layer H1 scenarios have

strict-monomial mean structure but non-Gaussian noise — Student-t(3); SD growing with the log prediction (0.3× to 1.7×); 2% point outliers (+8σ); 6% group outliers (+5σ); AR(1) ρ = 0.6 — rejections there are false alarms by construction and are never pooled into structural power. Grid: n □ {50, 200, 1000}; relative Gaussian noise σ □ {0, 0.5%, 1%, 3%, 5%} in log space; 500 seeds per cell via a seed rule embedding the grid position.

S2 — SRSD-Feynman tasks (32 tasks × 500 seeds). From the 120-equation registry, 32 tasks pass the positivity screen under official configurations: 13 strict-monomial H0 and 19 non-monomial H1. Each training fit (n □ {50, 200, 1000}) is accompanied by a 2,000-sample same-distribution test set and a far-field probe: the first regressor extrapolated ×3 beyond both training domain edges, scored against the closed-form truth in a numerically stable form.

S3 — Family-stratified Feynman equations (30 equations × 30 seeds). Structural family classification from symbolic form: exact power law (12 H0); fractional power (1 H0); exponential (3 H1); rational (6 H1); trigonometric (5 H1); plus not-applicable members with explicit reasons. Each H1 equation runs over relative noise {0.5%, 1%, 3%, 5%} × {homoskedastic, heteroskedastic} × sampling-range scale {official, ×0.5, ×2}; H0 equations run the noise grid at official range.

3.4 Suite S4 — the family-viability battery

Each test of §3.2 evaluates a single null form; a rejection terminates inquiry. The battery registers four families with their linearizing coordinate pairs — power (log x, log y); exponential (x, log y); logarithmic (log x, y); rational (1/x, 1/y) — and for each family F runs A's construction in F's coordinates, returning the viability set V = {F : p_F ≥ α} rather than a verdict. Columns are standardized before squaring so features at different scales weight equally, and the null is a wild bootstrap (Rademacher signs; B = 999) rather than a permutation — for the reason §4.6 documents: **null validity is a coordinate-dependent property**. The prediction (preregistered before any run) was that permutation, valid wherever the v-coordinate is log y, would break in coordinates ending in y or 1/y; the probe measured exactly that, and the formal run uses the wild null throughout. Three errors are reported separately and never merged: false exclusion (truth absent from V), false inclusion (wrong family present), and escalation on structures no family owns — the correct outcome, never scored as a win. Grid: n = 200; rel □ {0.5–5%}; 500 seeds; 188 cells; 94,000 datasets, all 19 S1 scenarios with truth-family labels frozen in code plus 14 preregistered SRSD H1 equations × {homoskedastic, heteroskedastic}.

3.5 Suite S5 — the gate comparison

S5 exists because of an objection this benchmark could not answer from its own data: *why not skip the test statistic entirely and let a nested rival arbitrate?* Gates E and F (§3.2) are that objection, made runnable. Design, frozen in `gate_compare.py`'s docstring before the run: benches = S1's 19 scenarios and S2's 32 tasks; grid n □ {50, 200}, rel = 3%, 200 seeds per cell (seed bases 300,000 / 400,000 — disjoint from all prior suites); rival = full quadratic expansion; E and F thresholds naive and H0-pool-calibrated per bench × n; A run under its standard protocol on identical data. Metrics: H0 size (naive and calibrated), structural-H1 detection at calibrated FPR 5%, noise-layer fire rates (should stay quiet), per-scenario and per-equation detail (never macro-averaged away), and cost in fit units. One asymmetry is inherent and is reported rather than hidden: A's permutation null is self-calibrating, while E/F at naive thresholds inherit nested-comparison tie bias; the calibrated variants price what a reference pool buys.

3.6 Suite S6 — real data (World Bank open macro data)

Synthetic and equation-derived suites have ground truth by construction; the deployment question is whether the synthetic failure-mode map predicts anything on observational data, where no truth exists. S6 is deliberately small, preregistered, and divergence-oriented: six macroeconomic relations (target, features) × four years (2019, 2021, 2022, 2023), cross-country cross-sections of 184–199 sovereign economies (World Bank aggregates excluded), from eight indicators pulled from the World Bank API with raw responses cached for reproducibility — life expectancy ~ GDP per capita (PPP); infant mortality ~ GDP per capita; GDP ~ population; CO_2 per capita ~ GDP per capita; internet penetration ~ GDP per capita; and life expectancy ~ GDP per capita + health expenditure per capita (two-feature). Gates A–D run at 5 permutation seeds per cell. These seeds are computational replicates of one p-value estimator on one fixed dataset — not five independent samples — so they are used only to report Monte Carlo stability (the empirical p has resolution 1/1000 at B = 999; across the 24 S6 cells the median p spread is .009 and the max .094), never as a sample-size or stability claim; a preregistered probe adds three agricultural and energy relations over three years, including a built-in control-contrast pair — cereal yield ~ fertilizer (one feature) versus cereal yield ~ fertilizer + energy use per capita (two features) — chosen to test, on live data, whether the dilution mechanism of S1 (a second

correlated curved feature halving the mean-square column's signal) operates outside the laboratory. No calibration or power claim is made on S6; its contract is external validity of *mechanisms*, not of rates. An engineering companion bench (Appendix E) extends the same contract to physical-engineering data: nine open sources, 91 datasets, 150 preregistered relations, run as a single-realization smoke pass.

4 Results

Table numbering is consecutive: Tables 1–2 size, Tables 3–4 power, Table 5 the family map, Table 6 noise, Table 7 extrapolation, Tables 8–10 the battery, Tables 11–12 the gate comparison, Tables 13–14 real data. All rates are rejection fractions; Wilson 95% intervals for headline rates appear in Appendix C. Seed counts per cell: 500 (S1, S2, S4), 30 (S3), 200 (S5), 5 (S6 — permutation-p stability, not rate estimation).

4.1 Size: false-positive rates on true monomials

Table 1 — S1 synthetic H0, false-positive rate at α = 0.05 (Gaussian noise pooled over σ □ {0.5%…5%}; 4,500 fits per cell).

Method	n=50	n=200	n=1000
A — permutation curvature	.047 [.041–.053]	.037 [.032–.043]	.041 [.036–.047]
B — RESET fitted F	.045 [.039–.051]	.041 [.036–.047]	.038 [.033–.044]
C — RESET fitted HC3	.048 [.042–.054]	.045 [.039–.051]	.038 [.033–.044]
D — RESET exog HC3	.054 [.048–.061]	.042 [.036–.048]	.040 [.034–.046]

Table 2 — S2 physics-derived monomial tasks (13 SRSD-Feynman equations; synthetic draws from real equations, not measurements), false-positive rate at α = 0.05 (26,000 fits per cell).

Method	n=50	n=200	n=1000
A	.049 [.046–.052]	.053 [.050–.055]	.052 [.049–.054]
B	.050 [.047–.053]	.050 [.047–.052]	.053 [.050–.055]
C	**.422 [.416–.428]**	**.417 [.411–.423]**	**.418 [.412–.424]**
D	**.564 [.558–.570]**	**.563 [.557–.569]**	**.565 [.559–.571]**

On synthetic Gaussian data, all four tests are calibrated (Table 1). On the physics-derived designs the picture splits (Table 2): A and B remain calibrated (.049–.053, flat in n); both HC3 variants break identically at every sample size — C at .417–.422, D at .563–.565, intervals that do not move from n = 50 to n = 1000. This is not a small-sample artifact that more data cures; it is a structural miscalibration of the robustness correction itself on near-collinear built columns (Section 4.4 rules out noise as the cause; the data in Table 2 are clean by construction — what differs is the *design*).

The pooled rates also conceal the failure's granularity, which matters for any claim of generality over "physical laws": at n = 200 the HC3 failure is equation-specific, not uniform. Five of the 13 tasks (i.34.27, i.38.12, ii.27.18, iii.12.43, iii.15.14) reject at .998–1.00 for C while the other 8 sit at .048–.060, essentially nominal. The Wilson intervals of Table 2 treat all 26,000 fits as independent draws; because seeds repeat within a fixed equation, the honest uncertainty unit for generalization over equations is the task. Task-level bootstrap (resampling the 13 equations; Appendix C) gives C a 95% interval of [.197, .707], and leave-one-equation-out moves its pooled rate over [.368, .448], against [.053, .054] for A. Two qualifications follow. First, "42–56% on physical laws" is a pooled summary of a bimodal phenomenon — near-nominal calibration on most equations, total failure on a specific subset — and the open question is what makes that subset special (Section 5, Q3, shows the answer tracks design conditioning). Second, an independent-implementation check run in revision (Section 5, Q3) reattributes the failure: a standard SVD-based HC3-Wald implementation (statsmodels) is nominal-compatible on the very tasks where our pinv-based path rejects at 1.00 (8/120 rejections = .067, Wilson [.034,.126] over 60 seeds × 2 tasks — sufficient to reject the old rate of 1.00, not precision evidence of exact 5% calibration), and the two implementations agree to ≥3 significant digits on well-conditioned tasks. Tables 2, 3, and 6 therefore report properties of the harness's HC3 path — a real, silent, deployment-relevant hazard, but no longer evidence against the HC3 correction itself.

4.2 Power: raw, then calibrated

Raw power at α = 0.05 on S2 H1 (n = 200, pooled over rel ≥ 0.5%): A .52, B .28, C .63, D .94. A naive reading crowns D. Table 2 already priced that crown. Calibrating every method to a common 5% empirical FPR reorders the field:

Table 3 — S2 power at matched empirical FPR 5%.

Method	n=50	n=200	n=1000
A	**.41**	**.51**	.59
B	.19	.28	.42
C	.05	.28	.41
D	.19	.37	**.65**

At deployment-relevant sample sizes (n ≤ 200), A delivers roughly twice B's calibrated power; at n = 1000, D shows the highest pooled power (.65). Pooled values, however, are means over only 19 H1 equations, and task-level paired bootstrap (resampling equations; 10,000 replicates at the Table 3 thresholds) shows the D-at-n=1000 advantage is not established cross-equationally: Δ(D−A) = −.224 [−.418, −.016] at n = 50 (A significantly better), −.147 [−.426, +.133] at n = 200, +.060 [−.218, +.333] at n = 1000 — every interval either favors A or crosses zero, and D beats A on 10–47% of tasks. Δ(D−B) = −.008/+.086/+.238 ([-.213,+.185]/[−.208,+.372]/[−.076,+.534]: all crossing zero); Δ(D−C) = +.136 [+.033,+.250] at n = 50 and +.252 [+.025,+.478] at n = 1000 (C being the retracted path, this compares within the path-specific family). The one paired difference that excludes zero in favor of a rival-family gate is Δ(F−A) = +.177 [+.054,+.323] at n = 50 and +.176 [+.056,+.320] at n = 200 (F wins 63–68% of tasks; F cannot be paired with D without integrating D into the S5 harness — outstanding). The defensible reading: D has the highest *average* calibrated power in the current task pool at n = 1000, not established cross-equation superiority; τ thresholds describe decision boundaries and are no substitute for these intervals. No method exceeds .65 — most of S2's H1 mass is hard for every test, which Section 4.3 decomposes by family.

Table 4 — S1 blind-spot cartography (raw α, n = 200, rel = 1%).

Scenario	A	B	C	D
cubic (standardized odd term)	.40	1.00	1.00	1.00
interaction ($lx_1 \cdot lx_2$ cross term)	.21	1.00	1.00	.10
opposing curvature (+c, −c, exchangeable)	**.02**	.98	.98	1.00
dilution (one curved variable of two)	1.00	.88	.87	1.00

Three mechanisms, each verified by construction: (i) A's statistic is the mean of squared centered columns — under opposing curvature on an exchangeable design that mean has exactly zero population signal; (ii) a quadratic-projection statistic loses power on odd-symmetric deviations; (iii) B's fitted-value powers are a local Taylor surrogate for any smooth alternative — saturating on interaction where A (.21) and D (.10) struggle — and, as S3 shows, the same surrogate property near-blinds B on rational laws.

4.3 Family-stratified results and sensitivity

Table 5 — S3 calibrated power by family.

Family	A	B	C	D
exponential	.11	.05	**.63**	.04
rational	**.66**	.14	.50	.66
trigonometric	**.70**	**.73**	.08	.05

B — near-perfect on synthetic quadratics — is nearly blind to rational laws (.13 raw, .14 calibrated): the fitted-value Taylor surrogate absorbs precisely the low-order curvature by which a saturating law departs from a power law. D's raw dominance on rational (.95) and trigonometric (1.00) is false-alarm budget, not power: calibrated, its trigonometric advantage collapses to .05. C is the exponential family's only reliable detector (.63) while being worthless on trigonometric (.08). A misses exponentials (.11) but is otherwise the most even performer. Every method's strength sits on another method's blind spot; Table 5 has no row-major winner. Sensitivity: scaling every sampling domain ×2 moves exactly one family-method cell by more than 15 points (rational × A: .66 → .84, the expected direction); ×0.5

moves none; heteroskedastic replications agree within .02 on every cell.

4.4 Noise-layer robustness

Table 6 — S1 noise-layer H1 (strict-monomial mean structure) rejection rates at α = 0.05 — false alarms attributable to the noise assumption (n = 200, rel = 1%, 500 seeds).

Scenario	A	B	C	D
Student-t(3) noise	.034	.036	.030	.030
variance growing with prediction	.048	.058	.050	.050
AR(1) errors, ρ = 0.6	.038	.040	.042	.042
2% point outliers (+8σ)	.040	.066	.036	.036
6% group outliers (+5σ)	.038	.060	.054	.054

Every method, every non-Gaussian regime: .030–.066. The table's function is exclusionary: the miscalibrations of Table 2 do not come from noisy data; the design, not the noise, is what specification tests are fragile to.

4.5 Do alarms predict extrapolation failure?

Base rate of far-field relative error above 10% across S2 fits: .198.

Table 7 — Alarm precision and recall for far-field error > 10% (S2, all tasks, all n, rel ≥ 0.5%).

Method	precision	recall
A	.39	.63
B	.56	.49

An A alarm raises the probability of severe extrapolation failure from .198 to .39 (2.0×); a B alarm to .56 (2.8×). Of the true-positive catches, 10,560 are A-only and 5,447 B-only against 12,575 shared — the alarm sets overlap barely more than chance. But alarms are ordinal evidence, not magnitude certificates: the suite's two worst extrapolators (i.39.11, i.43.43, both 98% median far-field error) carry the least in-domain curvature and are missed by every method, while ii.38.14 is rejected by A essentially always yet extrapolates benignly (7.7%). Alarm □ form evidence; silence □ nothing. No rejection is evidence of extrapolation safety.

4.6 The battery: from rejection to routing

Table 8 — Probe: permutation vs wild null by family (n = 200, 200 seeds; entries rel = 1% / 3%).

Truth	perm null	wild null
power (size of power member)	.070 / .075	.075 / .060
rational: false escalation (V = □)	**.34 / .31**	.045 / .090
rational: containment	.66 / .70	**.96 / .91**
exponential: containment	.94 / .95	.92 / .95
logarithmic: containment	.87 / .93	.89 / .95
threshold (no battery family): escalation — *correct outcome*	1.00 / 1.00	1.00 / 1.00

With the permutation null, the battery's own rational member rejects its own true family a third of the time — nothing about rational laws is at fault; the null is exchangeability-invalid in (1/x, 1/y) coordinates. The wild bootstrap repairs exactly that and nothing else.

Set-level errors and multiplicity. The battery's decision is the viable *set*, so the errors that matter are set-level, and each member runs at α = .05 without correction — a union decision whose two error types move oppositely under multiplicity. On H0-side cells the reported containment is one minus the exclusion rate of the true family's own member (.0496–.066 across the four truth families; the viable set is never empty), and per-member α does not inflate that rate because exclusion depends on one member only. Inflation appears on the other side: some wrong member survives alongside the truth in 23.3% of power-truth cells (near-neighbour overlap: exponential/logarithmic members retain power data) and 92% of logarithmic-truth cells (rational retains log data) — reported as near-neighbour overlap

rather than suppressed in the released tables. On the structural-H1 pool (72,000 cells whose truth is no battery member by design) some family is retained in 100% of cells: routing there means *which wrong family is modal* (Table 10), and a genuinely empty set never occurs — the battery is a candidate filtering stage whose surviving set the discovery layer must still rank, not a final selection. Downstream selection error after the battery is out of scope here and unmeasured.

Table 9 — Battery on synthetic truths (500 seeds/cell, wild null).

Truth family	Containment (truth □ V)	False escalation (V = □)
power — size, 8 scenario groups incl. all noise layers	—	.0496 [.046–.053] pooled; per-scenario .043–.054
exponential	.95	.05
logarithmic	.94	.005
rational	.93	.066 pooled (≤ .082 worst rel)

Table 10 — Battery on out-of-family structures (escalation is the *correct* outcome).

Scenario	Correct escalation	Modal false V
quadratic / threshold / omitted variable	1.00 / 1.00 / 1.00	—
dilution	.92	rational
cubic (odd term)	.24	power (.68)
interaction	.05	power (.95)
opposing curvature	**.00**	power+rational (.84)
collinear-quadratic	.03	rational (.61)

Table 10 is Table 4 re-expressed as routing errors: the power member inherits A's documented blind spots, so those structures stay in V under the power label instead of escalating. On the SRSD side the routing is reported descriptively (median |V| 1–2 for 11 of 14 equations; i.30.5 → {exponential} in 78%; ii.38.14 escalates on 67%), because surface-form family labels dissolve under examination — the battery correctly rejects Planck-type "exponentials" that no linearizable family owns.

4.7 The gate comparison: why not just compare against a nested rival?

Table 11 — S5 size on H0 pools (200 seeds/cell; naive and calibrated thresholds for E/F).

H0 pool	A	E naive	E cal.	F naive	F cal.
synth n=50 (600 fits)	.080	.212	.050	.132	.050
synth n=200 (600)	.040	.183	.050	.135	.050
SRSD n=50 (2,600)	.051	.167	.050	.086	.050
SRSD n=200 (2,600)	.038	.152	.050	.123	.050

Table 12 — S5 structural-H1 detection at calibrated FPR 5% (mean over tasks).

	synth n=50	synth n=200	SRSD n=50	SRSD n=200
A (permutation curvature)	.774	.782	.348	.493
E (CV-quad)	.929	.934	.462	.628
F (AICc-quad)	**.930**	**.940**	**.525**	**.669**
OR(A, E)	.937	.941	.547	.675

Three findings. **(a) The obvious alternative is the better default detector, and the margin is exactly A's blind zone.** The quadratic expansion carries per-feature squares *and cross-terms*, so INTERACT (A .24/.26) and OPPOSING (A .03/.02) are caught at 1.00 by both rival gates, and the two SRSD extrapolation-blind equations of §4.5 (i.39.11, i.43.43) are detected at .99–1.00. The direction set, not the null construction, was the binding constraint. CUBIC curvature defeats all three gates equally (~.3): odd curvature is near-orthogonal to every even-column design — the honest general form of the blind-zone lesson. **(b) The rival gates do not come calibrated.** Because the rival nests the null, ties and noise break toward escalation: naive thresholds false-alarm at .15–.21 (CV) and .09–.14 (AICc), and

calibration to 5% required an H0 reference pool — knowledge a production gate does not have. A's residual-permutation null is an approximation (§3.2) but was empirically size-valid on every pool we measured (.038–.080; synth n = 50 sits at .080 and is reported as measured) without any reference set — A's remaining structural advantage. **(c) Does the calibrated threshold transfer?** Calibrating E and F on all but one H0 equation and evaluating on the held-out equation (13-fold per bench × n) gives held-out sizes of .053/.050 (E, n = 50/200) and .051/.052 (F), with per-equation spread [.010, .095]: reference-pool calibration is not an in-sample artifact — it transfers across equations, though any single held-out equation can land within a ±.04 band of nominal. **(d) The OR union's recall advantage is a joint-budget artifact.** OR(A, E) with each gate at its own 5% runs a joint false-positive budget of .065–.102 across the four benches (Table 11) — its TPR edge over E (+.085/+.047 on SRSD) is purchased with that larger budget, not with information. Recalibrating the *combination* to a 5% joint budget (A's size fixed, E's threshold tightened on the H0 pool) collapses the union: at n = 50 A's own size consumes the budget and the union *is* A (TPR .348 SRSD / .774 synthetic); at n = 200 the recalibrated union reaches .664/.940 against F's .669/.940 — parity with F, never a gain. The corrected reading: at any common error budget F is the frontier; the union is defensible only when a workflow explicitly budgets for its larger naive size. Noise-layer scenarios leave all three gates quiet (.04–.07): the gates measure form, not noise assumptions, and the §4.4 separation survives its most obvious stress test. Cost in fit units: A ≈ 2 full fits + 1 vectorized pseudo-inverse (999 permutations); F = 2 fits; E = 12 — all negligible against the candidate flood a discovery pipeline already fits, which is the actual status quo's cost point: the workflow runs *none* of them today.

4.8 Real data: does the map predict live behavior?

Table 13 — S6 World Bank relations: verdicts (median permutation p over 5 seeds for A; deterministic p for B–D; α = .05).

Relation (target ~ features)	Years all four gates agree	Divergent cells (which gate, which way)
infant mortality ~ GDP pc	4/4 reject	—
CO_2 pc ~ GDP pc	4/4 reject	—
internet penetration ~ GDP pc	4/4 reject ($p < 10^{-16}$)	—
GDP ~ population	4/4 **quiet** (A .71–.91; B .19–.33)	—
life expectancy ~ GDP pc	0/4	A alone rejects 2019 (.035) & 2022 (.008); B alone rejects 2022 (.0002); C/D never
life expectancy ~ GDP pc + health exp pc	2/4 quiet	B rejects 2022 (.0002) while A quiet (.108)

Where signal is strong the gates agree — three relations rejected unanimously every year, one true scaling relation cleared unanimously — and A's five p-values per cell are Monte Carlo replicates of one estimator whose spread is small (median .009, max .094 at resolution 1/1000; no verdict flips beyond boundary cells). Divergence concentrates where the synthetic map puts weak curvature: the life-expectancy cells flip year to year (signal at the detection boundary), and the two-feature cell shows B rejecting where A stays quiet. We label this pattern *consistent with* the dilution mechanism (§4.2, Table 4) rather than a reproduction of it: on observational data the p-value change after adding a second feature can also reflect feature correlation, omitted-variable changes, sample composition, measurement error, cross-country heterogeneity, or spatial dependence — no ground truth exists to separate these, so the claim is signature-consistency, not mechanism confirmation.

Table 14 — S6 dilution probe: the same target with and without a second correlated curved feature (3 years × 5 seeds; detection = median gate verdict).

Relation	A	B	D
cereal yield ~ fertilizer (1 feature)	rejects 3/3 years (.011–.031)	rejects 3/3 (~.003)	rejects 3/3
cereal yield ~ fertilizer + energy pc (2 features)	**quiet 3/3 years** (.067–.149)	rejects 2/3 (.025–.092)	rejects 3/3 (≤.018)

This matches the S1 DILUTION scenario's signature — adding one correlated curved feature silences A on a relation it was rejecting, while the RESET variants continue to alarm — a pattern *consistent with* the dilution mechanism rather than a demonstrated reproduction of it, since feature correlation, sample composition, and omitted-variable changes move the same p-values in the same direction. No ground truth exists on S6 and none is claimed; the contract of §3.6 is

signature consistency in both directions: unanimous cells where the map predicts signal, and map-consistent divergence where it predicts weakness.

A companion engineering bench (Appendix E) pushes the same probe across nine physical-engineering sources (147 run cells): unanimous detection of a documented rating-curve misspecification, mechanism-echo divergences matching the synthetic map (small-sample power, interaction carried by a second feature, all three HC3 failure modes live), and mutual misses on locally approximate laws — all recorded, none adjudicated.

4.9 Runtime

Per-dataset wall time (S1 means over 500-seed cells): A 12.0 ms (n = 50) to 25.3 ms (n = 1000); B/C/D 1.0–1.8 ms; the full four-family battery at n = 200 runs in 24 ms; S5 gates E/F 1.5–3.5 ms. All gates are effectively free at the scale of a discovery pipeline that already fits thousands of candidates; the bottleneck resource is not compute but the decision discipline.

5 Analysis

Q1. Does the low-df curvature test beat RESET on univariate quadratic deviations? No — it ties; both saturate from n = 50. What low degrees of freedom buy is calibration, not power — which is what makes A's calibrated numbers trustworthy rather than large. The asymmetries that do appear are structural (odd terms, interactions), where A's quadratic projection is mathematically the wrong instrument.

Q2. Does averaging squared columns miss feature-specific curvature and interactions? Yes, by construction — and the benchmark quantifies when that matters, in three escalating venues: synthetic power cells (Table 4), routing errors (Table 10), and live observational data (Table 14). The dilution mechanism in particular is now a three-level result: constructed (S1), routed (S4), and signature-consistent on observational data (S6). A synthetic blind spot stops being a laboratory artifact the moment it anticipates which real relation goes quiet — though on S6 that correspondence remains pattern-level, not mechanism-proof.

Q3. Does HC3 fix false alarms under heteroskedasticity — and was the observed failure HC3's or ours? Heteroskedasticity was never the disease (Table 6). Table 2 shows *our* HC3-Wald path breaking on near-collinear built designs; the original draft attributed this to the correction itself ("the robust estimator is the failure mode"). The independent-software checks required in revision retract that attribution, and the second round extends them from fitted-HC3 (2 tasks × 60 seeds) to **exog-HC3 across all 13 H0 equations with stratified sampling by equation, sample size, and noise level (176 fits)**, on identical data frames, comparing three paths: our released pinv(X′X) path, our own SVD path (hat and sandwich via pinv(X) of the augmented design), and statsmodels HC3-Wald with the same F reference. Results: (a) the SVD path and statsmodels agree everywhere — maximum $|\Delta p| = .0023$, maximum relative statistic difference .0038, covariance-matrix Frobenius relative difference $\leq 3\times10^{-4}$ — so the mathematical definition is implemented identically and the divergence is numerical; (b) the pinv(X′X) path rejects at rate 1.00 on the 7 tasks whose measured cond(X′X) $\geq 4\times10^{17}$ (measured cond(X) 6.5×10^{8}–3.2×10^{10}), while both SVD paths stay at 0–.10 on every task including those — compatible with nominal, not precision-calibrated; (c) the two quiet tasks with cond(X) ≈ 10 (ii.27.16, ii.27.18) show no failure on any path. Mechanism, stated precisely: the harness calls NumPy 2.5.2 `np.linalg.pinv`, whose default cutoff zeroes singular values below rcond = 10^{-15} × σ_max(the max(m,n)·ε default belongs to SciPy's pinv, which we do not use); forming X′X squares the condition number, and per NumPy's own conditioning documentation a condition number 10^{k} loses ≈ k digits in the worst case — at measured cond(X′X) = 10^{17}–10^{20} the smallest singular values fall below the cutoff and are silently discarded, so the hat diagonal and sandwich are computed from a destroyed spectrum. The earlier draft's informal precision arithmetic is superseded by these measured values. There is no per-call rolling cross-check in the harness — the independent comparison is the released one-off script described above. Corrected conclusion: a silent numerical failure of pinv(X′X)-based inference on squared-condition-number designs; the practical hazard is real (research code commonly pinv-solves; the false alarms are total and silent), but it is an implementation robustness lesson, not evidence against HC3; whether any residual HC3 miscalibration survives elsewhere is open, and C/D rows in Tables 2, 3, and 6 should be read as path-specific outside the verified cells.

Q4. Do alarms predict extrapolation error? As binary evidence, usefully (2.0–2.8× enrichment, complementary alarm sets); as magnitudes or guarantees, no. The correct reading is asymmetric: alarm □ form evidence; silence □ nothing.

Q5. Can the same machinery answer "not a power law — then what?" Yes, as a viability set, after one repair the main benchmark could not have predicted: nulls are coordinate-dependent, and the wild bootstrap is the repair. The formal run delivers the contract (containment .93–.95, pooled size .0496, correct escalation 1.00 on quadratic/threshold/omitted-variable), with near-neighbour overlap reported rather than suppressed and A's blind spots wearing routing clothes (Table 10).

Q6. Why not just compare against a nested rival? Measured, it wins. F (AICc-quad) leads every cell of Table 12 and removes A's blind spots, because the rival's direction set (per-feature squares and cross-terms) covers the directions A's single mean-square column cannot project onto. The purchase price is a threshold: naive E/F false-alarm at .09–.21 until an H0 reference pool calibrates them; the calibration transfers across equations (leave-one-equation-out held-out sizes .050–.054 on average, §4.7(c)), while A was empirically size-valid on every pool we measured without one (§3.2 records the approximation this rests on). The OR combination, originally reported as the best-recall S5 cell, owes that edge to a larger joint false-positive budget (.065–.102); recalibrated to a 5% *joint* budget it matches but never beats F (§4.7(d)). The portfolio reading follows: a pipeline that can afford a reference pool should default to F; one that must report a p-value per candidate with no reference data should default to A; and a recall-critical deployment should budget for F rather than pay for an uncalibrated union. The benchmark's best test is thereby demoted from default gate to self-calibrating portfolio member, by the benchmark's own protocol; we consider this the paper's most credibility-relevant result, because it is the one a competing paper would have had an incentive to omit.

Q7. Does any of this transfer to real data? Where the synthetic map predicts strong signal, four gates built for physical laws agree on macroeconomic relations they were never designed for (Table 13) — cheap evidence that the map is not a laboratory artifact. Where the map predicts weakness, observational divergence appears with the predicted signature (dilution-consistency, Table 14) — consistent-with, not confirmation-of: too many observational confounds and no truth to adjudicate. What S6 and Appendix E cannot supply — rates, calibration, extrapolation truth, mechanism identification — they do not attempt; real data here is a compatibility probe, not a leaderboard, and their rejection counts are compatibility statistics, not performance rankings.

6 Limitations

Single candidate family (main benchmark). All gates of §3.2–3.5 evaluate a power law; the battery addresses routing, not selection — ranking within V is the discovery layer's job and is not benchmarked. **One rival, one criterion each (S5).** The rival is the quadratic expansion and the criteria are 5-fold CV and AICc — the reviewer's canonical objection, not a survey of selection gates; LOOCV, held-out likelihood, or non-nested rivals may rank differently, and the calibration asymmetry (reference pool vs self-calibrating null) is a workflow property whose cost we price but do not optimize. **Battery scope.** n = 200 only; no member for interaction or trigonometric structures; SRSD side descriptive pending a symbolically exact linearizability registry. **Real-data scope (S6).** Six relations, one source, four years, no ground truth: no rate claims are possible, the relation list is preregistered to prevent cherry-picking but remains six choices, and the year-to-year verdict flips in weak-curvature cells signal genuinely boundary-level signal rather than gate defect — these cells are reported as measured, not smoothed. **Fixed augmentation.** Built powers fixed at (2, 3). **Log-space scope.** All gates operate in log-log space; additive-error models are out of scope and marked not-applicable. **Statistical precision and clustering.** S1/S2/S4 rates rest on 500 seeds; S3 on 30 (directional); S5 on 200; S6's 5 seeds are Monte Carlo replicates of one estimator, not samples (median p spread .009, max .094). Seeds repeat within equations, so Table 2's Wilson intervals understate cross-equation uncertainty; task-level bootstrap intervals and leave-one-equation-out ranges (Appendix C) are the honest units for generalization over equations, and the HC3 failure is concentrated in 5 of 13 equations. No multiple-comparison correction is applied across the 1,480 cells; rates are descriptive, and family-level set decisions are analyzed as sets (§4.6) rather than as four independent tests. **HC3 independent verification.** After two revision rounds: fitted-HC3 (C) is verified on 2 failing tasks × 60 seeds; exog-HC3 (D) is verified across all 13 H0 equations with stratified sampling (176 fits) — in both cases our SVD path matches statsmodels to $\leq .0023$ in p and $\leq 3\times10^{-4}$ in covariance, while the $X'X$ path fails only at measured $\text{cond}(X'X) \geq 4\times10^{17}$. Outstanding: the same sweep across the remaining suites, noise layers, and the calibrated-power rows. **Permutation validity.** The residual-permutation null is an approximation whose finite-sample behavior under leverage, dependence, or stronger heteroskedasticity is unmeasured here; leverage-adjusted, wild, block, cluster, and fixed-design null variants are the required stress set. **OR budget.** Union gates are reported at both naive and joint-5% calibration; any deployment comparison must fix the budget first (§4.7(d)).

7 Reproducibility statement

Every generator is seeded from a documented rule embedding the grid position, so any cell of any table regenerates bit-identically. The harness records content hashes of the code under test — including the exact `form_check` implementation — and of the harness itself; a statistic-mismatch sentinel verifies on every call that the wrapper's recomputation agrees with the native implementation. The battery is implemented at the experiment layer over the unmodified core code path, so audited core-code hashes remain valid. S5's gate definitions and grid are frozen in `gate_compare.py`'s preregistration docstring (seed bases disjoint from all prior suites). S6 pulls from the World Bank API with every raw response cached under the run directory (`runs/wb_real/raw/`), making the observational suite reproducible against the cached snapshot and refreshable against the live API; the relation and probe lists are frozen in `wb_run.py` / `wb_blindspot.py` before any gate ran. The revision's re-analyses are scripted and released with their raw inputs: `paper/revision_analysis.py` regenerates every number in §4.1's task-level paragraph, §4.6's set-level errors, §4.7's OR-budget and transfer analyses, and §4.8's Monte-Carlo spreads from the released per-cell parquet/CSV files; `paper/verify_round2.py` regenerates the second-round D-verification, Wilson and paired-bootstrap numbers; the independent HC3 check (Q3) uses statsmodels 0.14 on identical data frames. The engineering companion bench of Appendix E is released the same way (`real_data/`): raw snapshots cached with SHA-256 hashes and access dates (2026-09-06), a manifest of preregistered relations with per-relation notes and per-source license fields, and replayable fetch/prepare/smoke scripts; every declined evaluation carries a typed reason. The benchmark artifact (repository archive with all generators, per-cell results, and content-hashed configurations) is deposited with the publication; a permanent DOI is assigned at release. All cells' raw results (parquet and CSV), failure registrations with typed reasons, selection and exclusion tables, analysis scripts, and figure sources are released. Zero-noise cells deliberately exercise the failure-registration path; the battery's 94,000 datasets and S5's 20,400 datasets contain zero data-layer failures, which is itself a registered outcome.

8 Conclusion

We built the falsification-side benchmark that model discovery lacks: six suites, 508,825 fitted datasets and 472,500 held-out evaluations, one protocol that puts every gate on a common basis, and full registration of every failure. Its deliverable is not a winning detector — the benchmark measured its own best test against the objection any reviewer would raise and watched it lose the head-to-head — but a priced menu. Its most striking claim did not survive its own revision: the 42–56% HC3 false alarms were a silent numerical failure of a pinv-based inference path on near-collinear built designs — an SVD-based implementation is calibrated exactly where it fails — a retraction that is itself the benchmark working as designed. Raw power rankings invert under matched false-positive rates, and invert again across families: the test that dominates synthetic benchmarks is blind to rational laws; the test blind to interactions saturates on them synthetically. The nested quadratic rival adjudicated by AICc is the strongest default gate on both benches — paying for it with a naive 9–14% false-alarm rate until a reference pool calibrates it (a calibration that transfers across equations) — while the permutation curvature gate was empirically size-valid on every pool measured without one, and the OR union's apparent recall premium is a joint-error budget artifact that vanishes at a common budget. The battery extends the machinery from rejection to routing, after a one-line null repair that carries the paper's most general lesson: a null distribution's validity is a coordinate-dependent property. And on macroeconomic data the map does what a map is for: it predicts where four gates will agree, and which gate goes quiet on which relation — a synthetic blind spot, observed in the wild. Benchmark papers are usually measured by the leaderboards they produce. This one is measured by the failures it makes visible — including, deliberately, its own — because in scientific discovery, the model you silently accept is the one that breaks your extrapolation.

References

[1] Ramsey, J. B. (1969). Tests for specification errors in classical linear least-squares regression analysis. *Journal of the Royal Statistical Society, Series B*, 31(2), 539–551.

[2] White, H. (1980). A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica*, 48(4), 817–838.

[3] MacKinnon, J. G., & White, H. (1985). Some heteroskedasticity-consistent covariance matrix estimators with improved finite sample properties. *Journal of Econometrics*, 29(3), 305–325.

[4] Long, J. S., & Ervin, L. H. (2000). Using heteroskedasticity consistent standard errors in the linear regression model. *The American Statistician*, 54(3), 217–224.

[5] La Cava, W., Orzechowski, P., Burlacu, B., de França, F. O., Virgolin, M., Jin, Y., Kommenda, M., & Moore, J. H. (2021). Contemporary symbolic regression methods and their relative performance. *NeurIPS Datasets and Benchmarks Track*.

[6] Mundhenk, T. N., Landajuela, M., Glatt, R., Santiago, C. P., faissol, D. M., et al. (2022). Rethinking symbolic regression datasets and benchmarks for scientific discovery. *arXiv:2206.10540*.

[7] Udrescu, S.-M., & Tegmark, M. (2020). AI Feynman: A physics-inspired method for symbolic regression. *Science Advances*, 6(16), eaay2631.

[8] Bakarji, J., Champion, K., Kutz, J. N., & Brunton, S. L. (2022). Discovering governing equations from partial measurements with deep delay autoencoders. *Nature Computational Science*.

[9] Cranmer, M. (2023). Interpretable machine learning for science with PySR and SymbolicRegression.jl. *SciPy Conference*.

[10] Liu, R. Y. (1988). Bootstrap procedures under some non-i.i.d. models. *Annals of Statistics*, 16(4), 1696–1708.

[11] Davidson, A. C., & Flachaire, E. (2008). The wild bootstrap, tamed at last. *Journal of Econometrics*, 146(1), 162–169.

[12] Burnham, K. P., & Anderson, D. R. (2002). *Model Selection and Multimodel Inference: A Practical Information-Theoretic Approach* (2nd ed.). Springer.

[13] Arlot, S., & Celisse, A. (2010). A survey of cross-validation procedures for model selection. *Statistics Surveys*, 4, 40–79.

[14] World Bank (2026). World Bank Open Data API, indicators NY.GDP.MKTP.PP.CD, NY.GDP.PCAP.PP.CD, SP.POP.TOTL, SP.DYN.LE00.IN, SP.DYN.IMRT.IN, EN.GHG.CO2.MT.CE.AR5, EN.GHG.ALL.MT.CE.AR5, SH.XPD.CHEX.PC.CD, IT.NET.USER.ZS, AG.YLD.CREL.KG, AG.CON.FERT.ZS, EG.USE.PCAP.KG.OE. https://api.worldbank.org (cached snapshot released with the benchmark).

[15] UCI Machine Learning Repository. archive.ics.uci.edu (datasets: auto_mpg, automobile, concrete, slump, energy_efficiency, yacht, ccpp, air_quality, gas_turbine, tetouan, steel). Accessed 2026-09-06; raw snapshots and SHA-256 hashes released with the benchmark; redistribution per each dataset's stated license. [16] NIST Statistical Reference Datasets (StRD). itl.nist.gov/div898/strd. Accessed 2026-09-06; public domain. [17] NIST Chemistry WebBook, SRD 69. webbook.nist.gov. Accessed 2026-09-06; public domain (U.S. Government work). [18] U.S. Geological Survey. National Water Information System (NWIS instantaneous values, 12 gauges, WY2023) and FDSN event service (global M ≥ 5.5, 2000–2024). earthquake.usgs.gov / waterdata.usgs.gov. Accessed 2026-09-06; public domain (U.S. Government work). [19] Open-Meteo historical weather API (ERA5 reanalysis), 10 wind sites, 2023. open-meteo.com, CC BY 4.0 (ERA5: Copernicus licence). Accessed 2026-09-06; reanalysis, not in-situ measurement. [20] OpenML. openml.org (datasets: ailerons, cpu_act, elusage). Accessed 2026-09-06; per-dataset licenses vary and are recorded in the released manifest. [21] U.S. Department of Energy, fueleconomy.gov vehicle fuel-economy data, model years 1984–2025. Accessed 2026-09-06; public domain (U.S. Government work). [22] RDatasets (Vincent Arel-Bundock), datasets: pressure, cars, rock, trees, attenu, housing, electricity, stormer, whiteside, cement, petrol, women, airquality, puromycin. vincentarelbundock.github.io/Rdatasets. Accessed 2026-09-06; licenses inherit from each source package (mostly GPL/BSD) and are recorded in the manifest.

Appendix A — Suite composition

Suite	Cells	Fits per cell	Total fits	Extra evaluations
S1 synthetic	285	500	142,500	—
S2 SRSD-Feynman	480	500	240,000	240,000 test sets; 232,500 extrapolation probes
S3 family-stratified	392	30	11,760	—
S4 battery	188	500	94,000	—
S5 gate comparison (v3)	102	200	20,400	—

Suite	Cells	Fits per cell	Total fits	Extra evaluations
S6 real data + probe (v3)	33	5	165	—
Total	**1,480**	—	**508,825**	**472,500**

Outside the seeded totals: the Appendix E engineering bench contributes 150 single-realization smoke cells (147 run), one fit per gate per cell.

Appendix B — Failure registration taxonomy

Reason	Meaning	Exercised by
`degenerate_zero_residual`	Zero-noise cell on exact ground truth; test undefined	S1/S2 σ = 0 cells
`rank_deficient`	Design matrix numerically rank-deficient	collinear designs
`insufficient_df`	n − k < 1 after augmentation	n = 50 multi-feature cells
`hc3_leverage_degenerate` / `hc3_singular_covariance`	HC3 undefined	degenerate built columns
`statistic_mismatch`	Wrapper statistic ≠ native statistic (drift sentinel)	continuous audit
`checked_false_*`	Native test declines to run (non-positive data, n < 12, non-finite)	not-applicable tasks
`runner_error` / generation failure	S5 SRSD-side generation exceptions	S5
indicator archived / no data	S6 indicator not published for a year	S6 (`EN.ATM.CO2E.KT` → AR5 replacement, logged)

All registered failures remain in the audit trail; denominators are documented per table. The battery's 94,000 and S5's 20,400 datasets contain zero data-layer failures.

Appendix C — Headline Wilson 95% intervals

S2 H0, n = 200 (26,000 fits per cell): A .053 [.050–.055]; B .050 [.047–.052]; C .417 [.411–.423]; D .563 [.557–.569]. S1 H0, n = 200 (4,500 fits per cell): A .037 [.032–.043]; B .041 [.036–.047]; C .045 [.039–.051]; D .042 [.036–.048]. Battery power-family size: .0496 [.046–.053]. S5 H0 pools (600–2,600 fits per cell): intervals within ±.013 of every Table 11 entry. Intervals for all other cells are computable from released per-cell counts.

Task-level uncertainty (S2, n = 200). Seeds repeat within an equation, so fit-count intervals understate cross-equation uncertainty; the equation is the honest unit for generalization over physical laws. Resampling the 13 H0 equations (10,000 bootstrap replicates): A .0535 [.051–.056], B .0497 [.048–.052], C .417 [.197–.707], D .563 [.276–.782]. Leave-one-equation-out pooled rates: A [.053, .054], B [.049, .050], C [.368, .448], D [.527, .606]. Per-equation rates (C): 5 equations at .998–1.00 (i.34.27, i.38.12, ii.27.18, iii.12.43, iii.15.14), the other 8 at .048–.060; no single equation contributes more than 18.5% of C's false alarms and its pooled rate remains .368 with the worst equation removed — widespread across the collinear subset, not one bad task. Macro and micro averages coincide here only because the design is balanced (2,000 fits per equation); on unbalanced pools they should be reported separately.

Appendix E — The real-engineering data bench (companion to S6)

S6 probes mechanism transfer on macroeconomic aggregates. The deployment domain these gates are actually built for is physical-engineering measurement — fluid, civil, hydraulic, energy, meteorological, seismic, chemical, and aerospace data, where power laws are the daily candidate models. This appendix documents a companion bench of that kind, run under the same contract as S6: real data has no ground truth, so the bench registers compatibility and divergence, not error rates; its cells are single realizations and stand outside Appendix A's seeded totals.

Composition and discipline. Nine open, license-free sources; 91 datasets; 150 candidate log-log relations preregistered in the released manifest with per-relation physical notes; 761,515 raw rows (Table E1). Raw bytes are cached (~115 MB) with SHA-256 hashes; cleaning is replayable and imputation-free — rows with missing or non-positive values are dropped and logged, and the few unit conversions are explicit in the manifest (e.g., Whiteside °C → K). Simulator datasets that violate the real-data discipline (a marine-engine and two wave-energy numerical

models) are excluded and listed, as are WebBook fluid pages below the 10-row rejection-detection floor; one gauge whose instantaneous series was never published is logged as an empty skip. ERA5 cells are marked as reanalysis, not direct measurement.

Table E1 — Composition by source.

Source	Datasets	Relations	Rows	Examples
USGS NWIS rating curves	12	12	442,269	discharge ~ gage height, 12 gauges
UCI	11	38	145,920	CCPP, Tetouan power, steel, yacht, concrete
EPA fueleconomy	1	4	48,624	CO_2 ~ MPG; combined MPG ~ displacement
Open-Meteo / ERA5	10	20	87,600	100 m ~ 10 m wind speed (shear); gust factor
OpenML	3	8	21,997	aileron position; CPU utilization
USGS FDSN	1	3	12,334	felt reports, stations, intensity ~ magnitude
RDatasets	14	26	1,349	cars, trees, puromycin, stormer, rock
NIST StRD	13	13	968	certified nonlinear calibration curves
NIST WebBook	26	26	454	phase-transition enthalpies ~ temperature
Total	**91**	**150**	**761,515**	—

Protocol. Each cell applies the §3.2 gates unchanged at $\alpha = 0.05$ — A with its B = 999 permutation p, B/C/D as registered — on the positivity-filtered relation, subsampled to $n \leq 800$ (seed 42) to bound permutation cost. The run is a smoke pass: one realization per cell, compatibility plus first verdicts, not a rate estimate. Of 150 cells, 147 run; three are registered `insufficient_n` by design (Bennett5 and Roszman1 have no positive rows after the log-space positivity filter; Danwood has n = 6) — the bench's honest-failure probes working as intended.

Table E2 — Smoke verdicts on the 147 run cells.

Gate	Rejections	Registered failures
A — permutation curvature	97/147 (.660)	10 declines, all `checked_false_small_n` ($n \leq 11$)
B — RESET fitted F	111/147 (.755)	0
C — RESET fitted HC3	108/147 (.735)	0
D — RESET exog HC3	94/147 (.639)	3: `insufficient_df` (n = 13, four features), `hc3_leverage_degenerate` (n = 546), `rank_deficient`

A and B agree on 123/147 cells (92 both reject, 31 both quiet; 5 A-only, 19 B-only). The failure columns are themselves informative: D's three declines are exactly the three HC3 failure modes of Table 2 and Appendix B, now on observational data; A's ten are uniform declines-to-run on $n \leq 11$ cells; B and C never fail. Nothing fails silently — every declined evaluation is registered with a typed reason.

Designed regimes. The bench's design document assigns expected regimes to 66 of the 150 cells before the smoke pass (Table E3). The assignment is a reading aid, not a scored test: the H0-side laws are approximate by design, and at n up to 800 an approximate law is a detectable null. The 12 rating-curve cells — whose registered form $Q \sim h$ omits the documented stage offset h_0 of the true rating law $Q = C(h - h_0)^n$, a mismatch noted in the manifest — are rejected unanimously by A and B alike (12/12): detection of a documented misspecification, not a false alarm. The five H0-side cells A retains are those with no designed departure mechanism — the braking-distance v^2 law, the tree allometry, the road-load relation — plus the two shear sites with the smallest fitted curvature.

Table E3 — Designed-regime cells, rejections at $\alpha = 0.05$.

Designed regime	Cells	A	B	C	D
Approximate power laws (H0-side)	30	25	27	23	23
Known non-power-law physics (H1-side)	36	20	24	24	14

(A's H1-side count includes its nine WebBook declines at $n \leq 11$ as non-rejections.) On the H1 side no gate sweeps the board: the misses concentrate in WebBook enthalpy cells whose measured temperature window keeps the local power-law approximation good — e.g., toluene $\Delta_{vap}H \sim T$: all four gates quiet at n = 26 — a mutual-miss cell

recorded as measured.

Table E4 — Named cells: mechanism echoes and divergences; p-values at α = 0.05 in A / B / C / D order, — = declined with registered reason.

Cell	A / B / C / D	Reading
Wind shear $v_{100} \sim v_{10}$, 10 ERA5 sites (n = 800 ea.)	8/10 / 10/10 / 7/10 / 7/10	B flags every site ($p \le 2\times10^{-4}$); A retains exactly the two smallest-curvature sites (.23, .41) — a graded response, not a threshold artifact
Gust factor ~ v_{10}, same sites (n = 800 ea.)	10/10 ×4	unanimous rejection at every site; no truth to adjudicate
puromycin rate ~ conc (Michaelis–Menten), n = 23	**.011** / .069 / .169 / .169	A detects the saturation curvature, B misses at α = .05 — Table 3's small-sample power gap, live
stormer Time ~ Viscosity + Wt (calibration law), n = 23	.001 / 7×10^{-4} / 9×10^{-3} / —	single-feature cells quiet (A .94, B .99); the second feature carries the interaction the law contains
EPA CO_2 ~ MPG (stoichiometric inverse, exponent −1), n = 800	.030 / 1.5×10^{-4} / .727 / .727	A and B reject on a near-exact physical law bent by fleet drive-mix curvature
EPA combined MPG ~ displacement (road load), n = 800	.386 / .596 / .599 / .599	designed power law passes
CCPP PE ~ V, n = 800	.642 / 8×10^{-4} / ≈0 / .023	proxy curvature: V correlates .81 with the strongly curved AT, so B/C's fitted-power columns inherit AT's curvature while A's V-only direction set cannot — the mirror of dilution
automobile horsepower ~ engine size, n = 203	.878 / 1.2×10^{-4} / .094 / .094	divergence under heteroskedastic body-style scatter
rock perm ~ area + peri, n = 48	.835 / 1×10^{-5} / 5.6×10^{-6} / .596	A/D quiet, B/C reject; no truth to adjudicate
housing price ~ 4 features, n = 546	.723 / .683 / ≈0 / —	D's leverage-degeneracy failure mode, live at n = 546, while A/B/C return verdicts

Scope. What the bench supports is what S6 supports: compatibility at scale (147/150 cells run across nine sources, zero silent failures), failure registration on observational data, and mechanism-echo cells whose directions match the synthetic map. Its per-gate rejection totals (97/111/108/94 of 147) are compatibility statistics on single realizations without ground truth — they rank nothing, and reading them as performance would repeat exactly the leaderboard confusion the benchmark exists to correct — A's calibrated conservatism (the shear sites, the road-load law, puromycin at n = 23), A's interaction and dilution behavior (stormer, CCPP), D's structural fragility (all three failure modes live). What it cannot support: rates, calibration, or rankings — single realizations, no ground truth, and time-ordered NWIS/Open-Meteo series whose permutation-independence assumption is an approximation marked in the manifest. All four p-values per cell are released (`real_data/smoke_results.csv`) with the manifest (`real_data/clean/manifest.json`, SHA-256 per source) and replayable fetch/prepare/smoke scripts.

Appendix G — Change notes (v2 → v5)

1. **Suite S5 added (gate comparison).** The question "why not just

compare against a nested rival?" is promoted from an anticipated objection to a measured result: AICc-quad leads all four cells of Table 12; the permutation gate is repositioned from best-calibrated test to self-calibrating portfolio member. The repositioning is reflected in the abstract ("Fifth"), Q6, §8, and the title's framing ("gates" rather than "tests").

2. **Suite S6 added (real data).** World Bank macro relations with a

preregistered list and a preregistered dilution probe; unanimous strong-signal cells and mechanism-consistent divergence, including a live reproduction of the DILUTION signature (Table 14). No rate claims.

3. **Totals updated.** 1,345 → 1,480 cells; 488,260 → 508,825 fitted

datasets.

4. **S3 raw table dropped from body** (Table 5 in v2) to keep the table

count within workshop limits; the raw family map remains in released per-cell data and the calibrated Table 5 carries the claim. v2's Q2 and the "external validity" sentence in Q2 are superseded by the three-level dilution result (constructed → routed → observed).

5. **References extended** [12–14] (AICc, CV survey, World Bank data).
6. **Post-v3 addition: Appendix E (real-engineering bench).** A

nine-source engineering companion to S6 — 91 datasets, 150 preregistered relations, 761,515 rows — run as a single-realization smoke pass whose cells stand outside the seeded totals (unchanged): unanimous detection of a documented rating-curve misspecification, echoes of the small-sample power gap and the interaction-by-second-feature structure, all three registered HC3 failure modes, and mutual misses on locally approximate laws.

7. **Major revision (v4) — reviewer response.** (a) The OR(A, E)

combination's recall advantage is re-analyzed at the combination level: its naive joint false-positive budget is .065–.102, and recalibrated to a 5% joint budget it matches but never beats F (§4.7(d)); "best cell" claims withdrawn. (b) Fit-count Wilson intervals are supplemented with equation-level bootstrap intervals, leave-one-equation-out ranges, and per-equation rates (§4.1, Appendix C); the HC3 failure is shown to concentrate in 5 of 13 equations. (c) Permutation-null claims conditionalized throughout: residual permutation is an approximation calibrated empirically on our pools, not exact by construction (§3.2, §6). (d) **The HC3 claim is retracted and re-attributed**: an independent statsmodels check shows our pinv-based HC3 path — not the HC3 correction — produces the 42–56% false alarms (1.00 vs .067 on the worst tasks; ≥3-digit agreement elsewhere); Tables 2/3/6 C/D rows are now labeled path-specific (Q3). (e) S5 calibration is validated out-of-sample by leave-one-equation-out threshold transfer (§4.7(c)). (f) Real-data claims downgraded to signature-consistency with confounds stated (§3.6, §4.8, Appendix E); S6's five seeds reframed as Monte Carlo replicates with measured spread. (g) Battery set-level errors and multiplicity reported (§4.6). (h) Related work extended to the wider specification-testing literature; references [15]–[22] split with access dates and per-source licenses; [8] description corrected; Table numbering restored to consecutive (v3 left a gap at 5).

8. **Second-round revision.** (a) Independent verification extended to

exog-HC3 (D): all 13 H0 equations, stratified sampling, three-path comparison — our SVD path matches statsmodels ($|\Delta p| \leq .0023$, covariance $\leq 3\times10^{-4}$) while the X′X path fails only at measured cond(X′X) $\geq 4\times10^{17}$; the numerics paragraph now states the exact call (NumPy 2.5.2 pinv, rcond = 10^{-15} relative) and replaces informal precision arithmetic with measured condition numbers (Q3). (b) The .067 re-verification restated as 8/120, Wilson [.034,.126] — "compatible with nominal", not precision calibration (Q3, §4.1). (c) Task-level paired bootstrap added for calibrated power: Δ(D−A), Δ(D−B), Δ(D−C) intervals with win fractions; "D strongest" softened to "highest average power in the current task pool" (§4.2, Appendix C). (d) Glyph coverage audited across the compiled PDF (zero uncovered characters; the reported missing-symbol boxes could not be reproduced). (e) S2 counts unchanged (32 tasks: 13 H0 + 19 H1; no task was removed; 1,480 cells and 508,825 datasets stand as audited).